

Microplastic Polymer Analysis Report

Report ID	68946a21-352e-484b-b192-a1a1a161cea1
Generated On	19 May 2026 15:44
Status	Completed
Sample ID	FTIR-4637

Prediction Results

Detected Polymer	PP
Confidence Score	0.87%
Model Used	finetuned_model

FTIR Spectral Analysis

- 1455 cm⁻¹ → CH bending
- 1375 cm⁻¹ → CH₃ symmetric deformation

Confidence Interpretation

■■■■■■■■■■ 0.87%

Scientific References

[1] Karian, H. Handbook of Polypropylene.

[2] Maier, C. Polypropylene: The Definitive User's Guide.

[3] Andrady, A. Plastics and environmental sustainability.

[4] Fotopoulou, K. Microplastics in marine systems.

[5] Zhang, J. FTIR identification of polypropylene.

AI Generated Summary

The analyzed sample was identified as Polypropylene (PP) with a confidence score of 0.87%. The FTIR spectral characteristics strongly support the presence of PP polymer chains. The detected CH bending and CH3 deformation peaks at 1455 cm⁻¹ and 1375 cm⁻¹ are characteristic of the aliphatic functional groups in PP. This polymer is commonly found in automotive parts, household items, and packaging materials. The analysis indicates a reliable classification suitable for environmental and material identification studies.